Лабораторная работа 8

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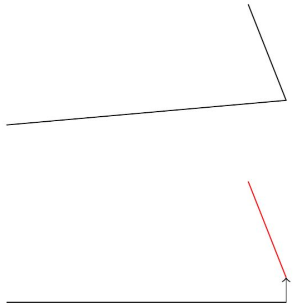
Начало работы с Tikz

```
\usepackage{tikz}
```

```
\begin{figure}
\begin{tikzpicture}
\draw (-1,0) -- (3,10pt) -- (35:3);
\end{tikzpicture}
\end{figure}

\begin{figure}
\begin{tikzpicture}
\draw[->] (-1,0) -| (3,10pt);
\draw[red] (3,10pt) -- (35:3);
\end{tikzpicture}
\end{figure}
```



Компоненты to и controls

```
\begin{figure}
\begin{tikzpicture}
\draw (-1,0) to (5,1);
\draw[green] (-1,0) to[out=90,in=135] (5,1);
\draw[cyan] (-1,0) .. controls (0,-2) .. (5,1);
\end{tikzpicture}
\end{figure}

\begin{figure}
\begin{tikzpicture}
\draw[dotted,gray] (-1,0) -- (5,1);
\draw (-1,0) .. controls (0,-2) and (4,2) .. (5,1);
\end{tikzpicture}
\end{figure}
```

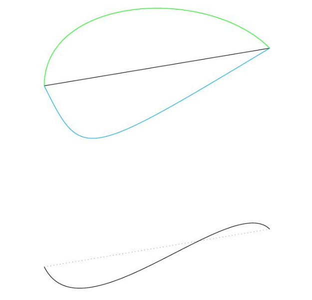
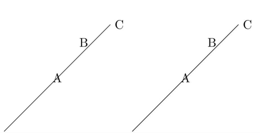


Рис. 1: to and controls

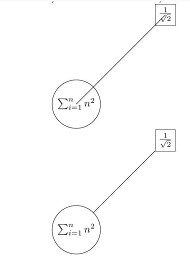
Подписи на иллюстрации

```
\begin{figure}
\begin{tikzpicture}[scale=3]
\draw (0,0) -- (1,1) node[midway]{A}
node[pos=0.75,above]{B} node[right]{C};
\end{tikzpicture}
\begin{tikzpicture}[scale=3]
\draw (0,0) to node[midway]{A} node[pos=0.75,
above]{B} (1,1) node[right]{C};
\end{tikzpicture}
```




```
\begin{tikzpicture}[scale=3]
\draw (0,0) node[circle, draw]{$\sum_{i=1}^{n}n^2$}
-- (1,1)
node[rectangle,draw]{$\frac{1}{\sqrt{2}}$};
\end{tikzpicture}

\begin{tikzpicture}[scale=3]
% define nodes
\node[circle,draw] (label1) at (0,0) {$\sum_{i=1}^{n}
n^2$};
\node[rectangle,draw] (label2) at (1,1) {$\frac{1}
{\sqrt{2}}$};
% draw the line
\draw (label1) -- (label2);
\end{tikzpicture}
\end{figure}
```



Ниже представлена илюстрация включающая в себя все функции о которых мы говорили ранее

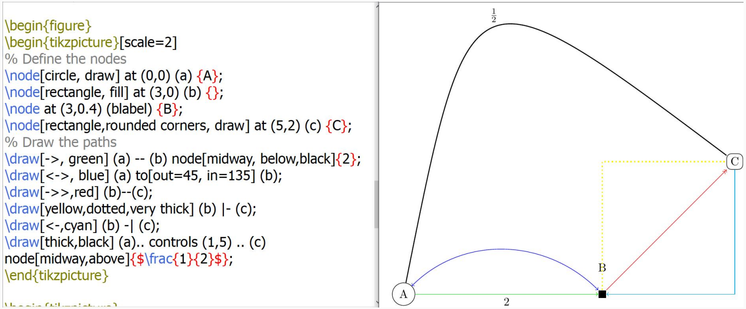
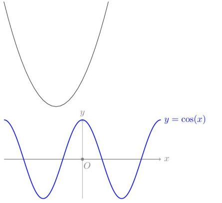


Рис. 2: example

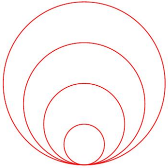
```
\begin{tikzpicture}
\draw [domain=-2:2] plot (\x, {pow(\x,2});
\end{tikzpicture}

\begin{tikzpicture}[scale=1.5]
% Draw the x and y axis, label the axes and the origin
\draw[gray, ->] (-2,0) -- (2,0) node[right]{$x$}
node[pos=0.53, below]{$O$}
\draw[gray, ->] (0,-1) -- (0,1) node[above]{$y$};
\draw[fill,gray] (0,0) circle [radius=1pt];
% Plot the curve
\draw[blue, thick] [domain=-2:2, samples=150] plot (\x,
{cos(pi*\x r)})
node[right]{$y = \cos(x)$};
% Note: the r in the argument of the cosine signifies that we
enter \x in radians
\end{tikzpicture}
\end{figure}
```




```
\begin{figure}
\begin{tikzpicture}[scale=0.75]
\foreach \x in {0,1,2,3}
\draw[red,thick] (0,\x) circle [radius=\x+1];
\end{tikzpicture}
\end{figure}

\end{document}
```



Практическая часть - Пример 1

```
\begin{tikzpicture}[scale=2]
% Define the nodes
\node[circle, draw, fill=green] at (0,3) (a) {A};
\node[circle, draw, fill=green] at (-3,-3) (c) {C};
\node[circle, draw, fill=green] at (3,-3) (e) {E};
\node[circle, draw] at (3,2) (f) {F};
\node[circle, draw] at (-3,2) (b) {B};
\node[circle, draw] at (0,-4) (d) {D};
% Draw the paths
\draw[blue,dotted,ultra thick] (b) -- (f) node[midway, above]
{6};
\draw[blue,dotted,thick] (b) -- (d) node[midway, above, left]
{2};
\draw[blue,dotted,thick] (d) -- (f) node[midway, above, right]
{4};
\draw[red,thick] (d) .. controls (3,-3) and (-3,2) .. (a)
node[midway, above, right]{$\sqrt{2}$};
\draw[red,thick] (a) to[out=-15, in=90] (e);
\draw[red,thick] (b) .. controls (-2,3) and (-1,4) .. (a);
\draw[cyan,ultra thick] (b) to[out=-110, in=110]  (c);
\draw[cyan,ultra thick] (c) to[out=-45, in=120] (d);
\end{tikzpicture}
```

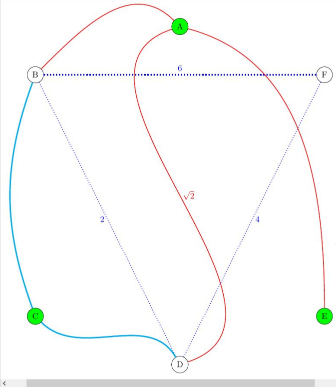


Рис. 3: exercise1

```
\begin{tikzpicture}[scale=2]
% Draw the x and y axis, label the axes and the
origin
\draw[gray, ->] (-1,0) -- (2,0) node[right]{$x$}
node[pos=0.33, below]{$O$};
\draw[gray, ->] (0,-2) -- (0,8) node[above]{$y$};
\draw[fill,gray] (0,0) circle [radius=1pt];
\draw (1,-0.05) -- (1,0.05) node[below, yshift=-2pt]
{$x=1$};

% Plot the curve
\draw[blue, thick] [domain=-1:2, samples=150] plot
(\x, {exp(\x)}) node[right]{$y = e^x$};
\draw[black, thick] [domain=0.1:2, samples=150]
plot (\x, {ln(\x)}) node[right]{$y = ln(x)$};
% Note: the r in the argument of the cosine signifies
that we enter \x in radians
\end{tikzpicture}
```

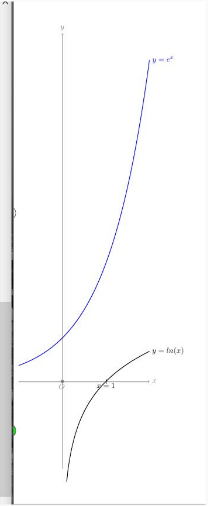
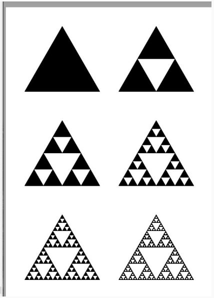


Рис. 4: exercise2

Фрактильные фигуры в Tikz

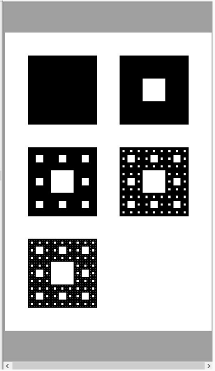
```latex
\documentclass[border=1cm]{standalone}
\usepackage{tikz}
\usetikzlibrary{math}
% Define a equilateral triangle with lower left corner at coordinate #1 and
% with length of the sides #2
\newcommand\Triangle[2]{
    \draw #1 coordinate(a) -- ++(0:#2) coordinate(b) ;
    \draw (a) -- ++(60:#2) coordinate(c);
    \fill (a) -- (b) -- (c) -- cycle;
}
\begin{document}
\begin{tikzpicture}
    \tikzmath{
    % Define the recursive function sierpinski
    function sierpinski(\x, \y, \s, \d) {
    if (\d == 0) then {
    % Draw a triangle lower left corner at (\x, \y),length \s
    { \Triangle{(\x,\y)}{\s}; };
        } else {
    % Rescale the length of the sides and choose correct coords
    % for the next triangles
    \u1 = 0.25*\s;
    \u2 = \u1*sqrt(3);
    \u3 = 0.5*\s;
    sierpinski(\x,\y,\u3,\d-1);
    sierpinski(\x+\u3,\y,\u3,\d-1);
    sierpinski(\x+\u1,\y+\u2,\u3,\d-1);
    };
};
% Let the length of the sides of the base triangle be 4, and generate 6 figures
\S = 4;
for \d in {0,...,5}{
% To situate all plots nicely under and next to each other, define the coords
% of the lower left corners preemptively
    \x = (\S+1)*mod(\d,2);
    \y = int(\d/2) * (\S+1);
    sierpinski(\x,-\y,\S,\d);
    };
}
\end{tikzpicture}
\end{document}
```



```latex
\documentclass[border=1cm]{standalone}
\usepackage{tikz}
\usetikzlibrary{math}

% Команда для квадрата
\newcommand\Square[2]{%
  \draw #1 rectangle ++(#2,#2);%
  \fill #1 rectangle ++(#2,#2);%
}

\begin{document}
\begin{tikzpicture}
\tikzmath{
  function carpet(\x,\y,\s,\d) {
    if (\d == 0) then {
      {{ \Square{(\x,\y)}{\s}; }};
    } else {
      \newS = \s/3;
      carpet(\x, \y+2*\newS, \newS, \d-1);
      carpet(\x+\newS, \y+2*\newS, \newS, \d-1);
      carpet(\x+2*\newS, \y+2*\newS, \newS, \d-1);
      carpet(\x, \y+\newS, \newS, \d-1);
      carpet(\x+2*\newS, \y+\newS, \newS, \d-1);
      carpet(\x, \y, \newS, \d-1);
      carpet(\x+\newS, \y, \newS, \d-1);
      carpet(\x+2*\newS, \y, \newS, \d-1);
    };
  };
  \S = 3;
  for \d in {0,...,4} {
    \x = (\S+1)*mod(\d,2);
    \y = int(\d/2)*(\S+1);
    carpet(\x,-\y,\S,\d);
  };
}
\end{tikzpicture}
\end{document}
```



Спасибо за внимание!